

Beyond Gradual Warming

European Food-and-Energy Exposure under CMIP6 Future Projections and Severe AMOC
Weakening Stress Tests

Thesis proposal presentation

May 2026

Motivation: why an AMOC stress test?

Climate signal

Recent evidence strengthens the case for studying severe AMOC weakening as a plausible stress-test pattern, even if hosing experiments are not forecasts.

- Portmann et al. (2026) estimate AMOC weakening of $51 \pm 8\%$ by 2100 under SSP2-4.5, stronger than the unconstrained CMIP6 multimodel mean. providing a more confident and stronger AMOC slowdown than the unconstrained estimate of $32 \pm 37\%$
- Xing et al. (2026) identify a two-decade decline in the observed western-boundary contribution to AMOC.
- van Westen and Dijkstra (2026) show that abrupt Gulf Stream path changes can act as an early-warning signal in a high-resolution model.

Economic question

These studies do not directly estimate crop or household-energy impacts. This thesis translates physical AMOC scenarios into country-level exposure metrics for Europe.

Main research question

Research question

To what extent would severe AMOC weakening generate European agricultural and household-energy exposure beyond the risks implied by standard CMIP6-based future climate projections?

Sub-questions

1. How have wheat and barley yields historically responded to crop-season weather?
2. How has household gas and electricity consumption historically responded to heating and cooling degree days?
3. What exposure is implied by standard ISIMIP3b future projections?
4. What additional exposure is implied by applying NAHosMIP AMOC anomalies to those projections?
5. Which countries show the largest difference between the two scenario pathways?

Hypotheses

H1: AMOC stress differs from standard projections

The AMOC stress-test scenario produces a different crop-energy exposure profile than standard ISIMIP3b projections.

H2: Northern and north-western countries are more exposed

Ireland, the United Kingdom, the Netherlands, Denmark, Norway, and Sweden are expected to show relatively stronger AMOC-related changes in winter heating demand and crop-season weather.

H3: Food and energy channels can move together

Lower crop-season heat accumulation or higher frost exposure may coincide with higher heating demand, creating combined food-energy stress.

Contribution and gap

Contribution

A public-data, multi-country framework that compares standard future climate exposure with severe AMOC-specific stress-test exposure for European crops and household energy demand.

Adjacent literature	Closest papers	Difference here
AMOC physical climate impacts	Jackson et al. (2023); Bellomo et al. (2023); van Westen et al. (2025)	Physical climate response; not country-level food-energy exposure. Great Britain and agriculture only; no household energy module. Integrated-assessment welfare model; not sectoral response functions.
AMOC agriculture	Ritchie et al. (2020)	
AMOC economics / IAMs	Link and Tol (2011)	
European energy demand	Wenz et al. (2017); De Cian and Sue Wing (2019)	Gradual warming and energy; not AMOC stress plus agriculture.

Novel element: crops plus household energy, country-level exposure, and a direct comparison of ISIMIP projections with AMOC-stressed weather.

Empirical scope

Dimension	Scope
Countries	EU27, Norway, the United Kingdom, and Switzerland; micro-states excluded.
Crop outcomes	Wheat and barley; dependent variable is FAOSTAT country-crop-year yield.
Energy outcomes	Household natural gas and household electricity; dependent variable is Eurostat country-fuel-year energy use.
Historical weather	E-OBS daily gridded temperature and precipitation.
Historical periods	Agriculture: as far as FAOSTAT and E-OBS overlap; energy: as far as Eurostat household energy, prices, population, and E-OBS overlap.
Scenario periods	ISIMIP3b future periods such as 2031–2060 and 2071–2100; AMOC stress from NAHosMIP monthly anomalies.

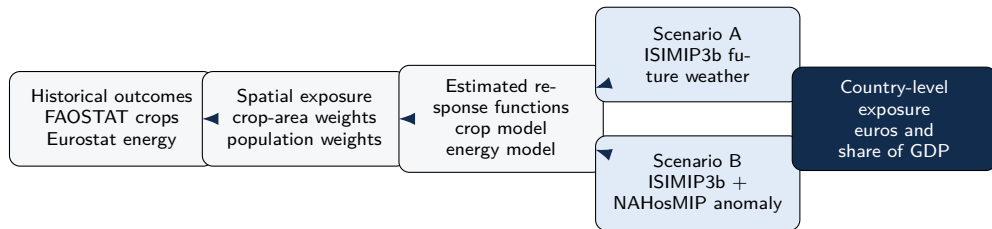
Key design principle

Subnational grids improve weather exposure measurement, but the final econometric units remain country-crop-year for crops and country-fuel-year for energy.

Input datasets and role in the workflow

Purpose	Dataset	Variables / role
Country outcomes	FAOSTAT	Crop yield, production, harvested area, producer prices.
Subnational crop geography	Harmonized EU subnational crop statistics	Regional crop area used only to build fixed wheat/barley weights.
Household energy outcomes	Eurostat nrg_bal_c	Household natural gas and electricity use in ktoe.
Energy controls	Eurostat	Fuel prices and annual population.
Historical weather	E-OBS	Daily mean, minimum, maximum temperature and precipitation.
Spatial geography	GISCO NUTS polygons; Census grid 2021	NUTS geometry and fixed population weights.
Future weather	ISIMIP3b	Daily tas, tasmin, tasmax, pr.
AMOC anomaly	NAHosMIP and piControl	Monthly AMOC temperature anomalies and precipitation ratios.

Overall architecture



Core sentence

Crop exposure is weighted by where wheat and barley are grown; energy exposure is weighted by where people live. Final regressions remain at country level.

Shared preparation: one country key and one spatial CRS

Country crosswalk

All datasets are harmonized to `country_id`. Raw country names are not used as merge keys.

country_id	country_name	faostat_area	eurostat_geo
FRA	France	France	FR
DEU	Germany	Germany	DE
ITA	Italy	Italy	IT
GBR	United Kingdom	United Kingdom	UK

`0.country_crosswalk.csv`

Project geography

All spatial overlays are done in **EPSG:3035**, the European equal-area projection.

Used for:

- NUTS polygons;
- E-OBS and ISIMIP grid-cell polygons;
- population grid;
- area-overlap weights.

Methodological choices now resolved

Previous concern	Resolved design
Use Eurostat HDD/CDD or compute them?	Compute HDD/CDD directly from E-OBS so the same method can be applied to ISIMIP and AMOC-stressed weather. Use Eurostat HDD/CDD only for validation.
Annual energy data but seasonal weather?	Keep annual Eurostat outcomes. Build two weather timings: Oct–Mar heating-season HDD and calendar-year HDD. Report one as main and the other as robustness.
Country-average weather or regional production geography?	Use subnational crop-area weights for crop weather and population weights for energy weather.
Extrapolation outside historical weather?	Flag every future scenario indicator against the historical 5th–95th, 1st–99th, and min–max range.

Crop module: outcome panel stays national

FAOSTAT crop outcome panel

The dependent variable remains national crop yield. Regional crop data only improve the weather exposure measure.

Item	Choice
Unit	Country $\times$ crop $\times$ year.
Crops	Wheat and barley.
Variables	Yield, production, harvested area; producer prices for valuation where available.
Output	<code>1.FAOSTAT_prepared.csv</code> : country_id, crop_id, year, yield_kg_ha, production_t, area_ha.

Crop module: fixed crop-area weights

Role of harmonized subnational crop statistics

Use the harmonized EU subnational crop dataset only to identify where wheat and barley are grown. Do not replace FAOSTAT outcomes.

Clean regional crop area

- Keep nuts_id, country_id, crop_id, year, area_ha.
- Match the NUTS polygon version to the crop statistics version.
- Output: 2.crop_subnational_area_clean.csv .

3.nuts_crop_area_weights.csv includes quality_flag: observed baseline, partial baseline, or fallback.

Baseline crop weights

$$Area_{r,c,k}^{base} = \frac{1}{|B|} \sum_{t \in B} Area_{r,c,k,t}$$

$$w_{r,c,k}^{crop} = \frac{Area_{r,c,k}^{base}}{\sum_{j \in c} Area_{j,c,k}^{base}}$$

Main baseline: 2010–2020. Robustness: 2000–2020.

Crop module: weather exposure from grid to NUTS

Spatial objects

- `4.nuts_polygons_clean.gpkg`
- `5.eobs_grid_polygons.gpkg`

E-OBS must be represented by grid-cell polygons, not only grid points.

E-OBS-to-NUTS crosswalk

For grid cell g and NUTS region r :

$$\phi_{g,r} = \frac{Area(g \cap r)}{\sum_h Area(h \cap r)}$$

$$\sum_g \phi_{g,r} = 1 \quad \forall r$$

`6.eobs_to_nuts_crosswalk.csv` : `eobs_cell_id`, `nuts_id`, `country_id`, `overlap_area`, `nuts_weight`

Crop module: daily indicators and aggregation

Daily grid indicators

For each E-OBS grid cell g and day d :

$$GDD_{g,d} = \max(tg_{g,d} - 5, 0)$$

$$Heat_{g,d} = \max(tx_{g,d} - 28, 0)$$

$$Frost_{g,d} = \mathbf{1}(tn_{g,d} < 0), \quad Precip_{g,d} = rr_{g,d}$$

Do not aggregate temperature first.

`7.eobs_nuts_crop_weather.csv`

NUTS-year-crop weather

For crop window S_k :

$$GDD_{r,k,t} = \sum_{d \in S_k} \sum_g \phi_{g,r} GDD_{g,d}$$

growing degree days

Same aggregation for heat, frost, and precipitation.

Main crop window: March–July. Robustness:
April–August.

Crop module: NUTS weather to country-crop-year

Production-weighted country exposure

Merge NUTS weather with fixed crop-area weights and aggregate to the FAOSTAT unit.

$$GDD_{c,k,t} = \sum_{r \in c} w_{r,c,k}^{crop} GDD_{r,k,t}$$

$Heat_{c,k,t}$, $Frost_{c,k,t}$, $Precip_{c,k,t}$ computed analogously.

```
8.eobs_country_crop_weather_weighted.csv
```

Final crop estimation panel

```
1.FAOSTAT_prepared.csv + 8.eobs_country_crop_weather_weighted.csv + crop prices.
```

Output: 9.crop_panel_estimation.csv .

Crop response function

Panel model

$$\ln(Yield_{c,k,t}) = \beta_{1k}GDD_{c,k,t} + \beta_{2k}Heat_{c,k,t} + \beta_{3k}Frost_{c,k,t} \\ + \beta_{4k}Precip_{c,k,t} + \beta_{5k}Precip_{c,k,t}^2 + \alpha_{c,k} + \lambda_t + \epsilon_{c,k,t}.$$

- Crop-specific weather coefficients allow wheat and barley to respond differently.
- Country-crop fixed effects absorb time-invariant productivity and climate differences.
- Year fixed effects absorb common shocks such as technology trends and Europe-wide market conditions.
- Interpretation: short-run historical weather response, not full adaptation.

Empirical motivation: Schlenker and Roberts (2009); Lobell and Burke (2010); Blanc and Schlenker (2017).

Energy module: outcome panel stays national

Eurostat household energy panel

The dependent variable remains annual household consumption by country and fuel. Population grids only improve the weather exposure measure.

Item	Choice
Unit	Country $\times$ fuel $\times$ year.
Dataset	Eurostat complete energy balances, <code>nrg_bal_c</code> .
Fuels	Household natural gas and household electricity.
Dependent variables	$\ln(\text{Energy})$ and, preferably, $\ln(\text{Energy}/\text{Population})$ as a cleaner specification.
Output	<code>10.energy_panel_raw.csv</code> : <code>country_id</code> , <code>fuel_id</code> , <code>year</code> , <code>energy_use_household_ktoe</code> .

Energy module: population-weighted exposure

Population grid

Use the Census grid 2021 in EPSG:3035 as fixed baseline household geography.

`11.population_grid_clean.gpkg`

Do not make population geography time-varying in the main specification.

E-OBS population weights

Overlay population cells with E-OBS grid-cell polygons.

$$w_{g,c}^{pop} = \frac{Population_{g,c}}{\sum_{h \in c} Population_{h,c}}$$

$$\sum_{g \in c} w_{g,c}^{pop} = 1 \quad \forall c$$

`12.eobs_population_weights.csv`

Interpretation

Energy weather is weighted by where households live, not by agricultural land area or simple country averages.

Energy module: daily HDD and CDD

Heating degree days

For each E-OBS grid cell and day:

$$HDD_{g,d} = \mathbf{1}(T_{g,d} < 15)(18 - T_{g,d})$$

Two annual measures are created:

$$HDD_{c,t}^{OctMar} = \sum_{d \in Oct_{t-1}:Mar_t} \sum_g w_{g,c}^{pop} HDD_{g,d}$$

$$HDD_{c,t}^{Calendar} = \sum_{d \in Jan_t:Dec_t} \sum_g w_{g,c}^{pop} HDD_{g,d}$$

Cooling degree days

For each E-OBS grid cell and day:

$$CDD_{g,d} = \mathbf{1}(T_{g,d} \geq 24)(T_{g,d} - 21)$$

Summer cooling measure:

$$CDD_{c,t}^{JJA} = \sum_{d \in Jun_t:Aug_t} \sum_g w_{g,c}^{pop} CDD_{g,d}$$

Output:

13.eobs_country_energy_weather_weighted.csv

Energy controls and estimation panel

Prices

14.energy_prices_prepared.csv

- country_id
- fuel_id
- year
- price, ln_price

Population controls

15.population_country_year.csv

- country_id
- year
- population
- ln_population

Final energy estimation panel

Merge energy outcomes, population-weighted HDD/CDD, fuel prices, and annual population.

16.energy_panel_estimation.csv

Energy response functions

Gas model: heating-driven

$$\ln(Gas_{c,t}) = \beta_H HDD_{c,t} + \beta_P \ln(GasPrice_{c,t}) + \alpha_c + \lambda_t + \epsilon_{c,t}$$

Use either HDD^{OctMar} or $HDD^{Calendar}$ as the main specification; report the other as robustness.

Electricity model: heating and cooling

$$\ln(Electricity_{c,t}) = \beta_H HDD_{c,t} + \beta_C CDD_{c,t}^{JJA} + \beta_P \ln(ElecPrice_{c,t}) + \alpha_c + \lambda_t + \epsilon_{c,t}$$

Empirical motivation: Auffhammer and Mansur (2014); De Cian and Sue Wing (2019); Wenz et al. (2017).

Scenario A: standard ISIMIP3b future projections

Purpose

Measure food-energy exposure under standard CMIP6-based future projections before adding any AMOC-specific stress signal.

Inputs

- ISIMIP3b daily climate input data.
- Variables: `tas`, `tasmin`, `tasmax`, `pr`.
- Scenario periods: 2031–2060 and 2071–2100, by SSP and climate model.

Processing

- Build ISIMIP grid-cell polygons in EPSG:3035.
- Create ISIMIP-to-NUTS and ISIMIP-to-population crosswalks.
- Compute the same crop and energy indicators as in the historical E-OBS workflow.

`17.isimip_country_crop_weather_weighted.csv`

`18.isimip_country_energy_weather_weighted.csv`

Scenario B: NAHosMIP AMOC anomaly

Idea

Construct a monthly AMOC stress anomaly from NAHosMIP and apply it to daily ISIMIP future weather. The AMOC experiment is a severe stress test, not a forecast.

Monthly temperature anomalies

$$\Delta tas_m = \overline{tas}_m^{u03-hos} - \overline{tas}_m^{picon}$$

$$\Delta tasmin_m = \overline{tasmin}_m^{u03-hos} - \overline{tasmin}_m^{picon}$$

$$\Delta tasmax_m = \overline{tasmax}_m^{u03-hos} - \overline{tasmax}_m^{picon}$$

Monthly precipitation ratio

$$R_m^{pr} = \frac{\overline{pr}_m^{u03-hos}}{\overline{pr}_m^{picon}}$$

Main experiment: u03-hos. Sensitivities can use alternative hosing levels and available NAHosMIP model members.

Scenario B: AMOC-stressed daily future weather

Apply monthly anomaly to daily ISIMIP weather

For each daily ISIMIP value, use the NAHosMIP anomaly for that calendar month.

$$\begin{aligned}tas_d^{AMOC} &= tas_d^{ISIMIP} + \Delta tas_{m(d)} \\tasmin_d^{AMOC} &= tasmin_d^{ISIMIP} + \Delta tasmin_{m(d)} \\tasmax_d^{AMOC} &= tasmax_d^{ISIMIP} + \Delta tasmax_{m(d)} \\pr_d^{AMOC} &= pr_d^{ISIMIP} \times R_{m(d)}^{pr}\end{aligned}$$

19.amoc_stressed_country_crop_weather_weighted.csv

20.amoc_stressed_country_energy_weather_weighted.csv

Scenario aggregation must match historical aggregation

Essential rule

The same spatial weights, crop windows, temperature thresholds, precipitation treatment, and annual timing rules used in historical estimation are also used for ISIMIP and AMOC-stressed weather.

Component	Historical estimation	Scenario projection
Crop exposure	E-OBS grid → NUTS → crop-area-weighted country indicators.	ISIMIP/AMOC grid → NUTS → same crop-area weights.
Energy exposure	E-OBS grid → population-weighted country HDD/CDD.	ISIMIP/AMOC grid → same population-weighting logic.
Outcome unit	FAOSTAT country-crop-year; Eurostat country-fuel-year.	Predictions reported by country, crop/fuel, scenario, period, model.

Prediction: crop impacts

Crop prediction equation

For each country c , crop k , scenario s , and future period:

$$\Delta \ln Yield_{c,k,s} = \hat{\beta}_{1k} \Delta GDD_{c,k,s} + \hat{\beta}_{2k} \Delta Heat_{c,k,s} + \hat{\beta}_{3k} \Delta Frost_{c,k,s} + \hat{\beta}_{4k} \Delta Precip_{c,k,s} + \hat{\beta}_{5k} \Delta Precip_{c,k,s}^2$$

- Baseline can be recent historical climate, for example 1991–2020.
- Approximate yield percentage change: $\% \Delta Yield \approx 100 \times \Delta \ln Yield$.
- Compare Scenario A and Scenario B using the same response coefficients.

21.crop_predictions.csv

Prediction: energy impacts

Gas

$$\Delta \ln Gas_{c,s} = \hat{\beta}_H \Delta HDD_{c,s}$$

Gas is mainly heating-driven, so the AMOC cooling channel is central.

Electricity

$$\Delta \ln Electricity_{c,s} = \hat{\beta}_H \Delta HDD_{c,s} + \hat{\beta}_C \Delta CDD_{c,s}$$

Electricity can respond to both heating and cooling.

22.energy_predictions.csv

Valuation and ranking

Crop exposure

$$CropExposure_{c,k} = \Delta Yield_{c,k} \times Area_{c,k}^{base} \times Price_{c,k}^{base}$$

Energy exposure

$$EnergyExposure_{c,f} = \Delta EnergyDemand_{c,f} \times EnergyPrice_{c,f}^{base}$$

Total exposure

$$TotalExposure_c = \sum_k CropExposure_{c,k} + \sum_f EnergyExposure_{c,f}$$

Final comparison

Rank countries by exposure in euros, exposure as a share of GDP, and the incremental difference between AMOC-stressed outcomes and standard ISIMIP outcomes.

```
23.total_exposure_country.csv
```

Expected result tables

Scenario comparison by country

Country	ISIMIP crop	AMOC crop	ISIMIP energy	AMOC energy	AMOC minus ISIMIP
Ireland	...	...	...	...	...
Germany	...	...	...	...	...
Norway	...	...	...	...	...
Spain	...	...	...	...	...

Main thesis output

- Which countries are most exposed under standard future projections?
- Which countries are most exposed under AMOC stress?
- How much does AMOC stress change the ranking and magnitude of exposure?

Validation checks

Crop validation

- Check that $\sum_{r \in c} w_{r,c,k}^{crop} = 1$.
- Report coverage of observed harmonized crop statistics.
- Compare crop-area-weighted weather against simple country-average weather.

`crop_weight_quality_report.csv`

Energy validation

- Check that $\sum_{g \in c} w_{g,c}^{pop} = 1$.
- Compare E-OBS calendar HDD/CDD with Eurostat HDD/CDD.
- Expect high correlation, not exact equality, because weighting and windows differ.

`energy_weather_validation.csv`

Historical-support flags

Why this matters

The response functions are estimated from historical weather variation. Scenario indicators outside the historical range imply extrapolation rather than interpolation.

Flag	Meaning for each future indicator
Green	Inside the historical 5th–95th percentile.
Amber	Outside 5th–95th but inside 1st–99th percentile.
Red	Outside 1st–99th percentile.
Black	Outside the historical minimum–maximum range.

`24.historical_support_flags.csv`

Robustness checks

Dimension	Robustness design
Crop weights	Main: 2010–2020 crop-area weights. Robustness: 2000–2020 and simple country-average weather.
Crop season	Main: March–July. Robustness: April–August.
Energy timing	Main: heating-season HDD or calendar HDD. Robustness: report the alternative.
Energy outcome	Main: gas and electricity separately. Robustness: total household final energy and per-capita specifications.
Scenario uncertainty	Report individual ISIMIP and NAHosMIP model combinations before reporting ensemble summaries.
Precipitation anomaly	Main: multiplicative precipitation ratio. Robustness: additive monthly precipitation anomaly.
Historical support	Report green/amber/red/black flags for every crop and energy indicator.

Implementation order

1. Build `0.country_crosswalk.csv` and enforce `country_id` everywhere.
2. Prepare FAOSTAT crop outcomes and Eurostat household energy outcomes.
3. Clean harmonized subnational crop areas and build fixed NUTS crop-area weights.
4. Load NUTS polygons matching the crop statistics version.
5. Build E-OBS grid-cell polygons and E-OBS-to-NUTS crosswalk.
6. Create E-OBS crop-weather and energy-weather weighted tables.
7. Estimate historical crop and energy response functions.
8. Only then process ISIMIP and AMOC-stressed weather using the same aggregation rules.

Practical rule

Finish the historical E-OBS pipeline first. Scenario processing should be a replay of the same pipeline with ISIMIP and AMOC-stressed weather.

Final file structure

Preparation and historical estimation

- 0.country_crosswalk.csv
- 1.FAOSTAT_prepared.csv
- 2.crop_subnational_area_clean.csv
- 3.nuts_crop_area_weights.csv
- 4.nuts_polygons_clean.gpkg
- 5.eobs_grid_polygons.gpkg
- 6.eobs_to_nuts_crosswalk.csv
- 7.eobs_nuts_crop_weather.csv
- 8.eobs_country_crop_weather_weighted.csv
- 9.crop_panel_estimation.csv
- 10.energy_panel_raw.csv
- 11.population_grid_clean.gpkg
- 12.eobs_population_weights.csv
- 13.eobs_country_energy_weather_weighted.csv
- 14.energy_prices_prepared.csv
- 15.population_country_year.csv
- 16.energy_panel_estimation.csv

Scenarios, prediction, validation

- 17.isimip_country_crop_weather_weighted.csv
- 18.isimip_country_energy_weather_weighted.csv
- 19.amoc_stressed_country_crop_weather_weighted.csv
- 20.amoc_stressed_country_energy_weather_weighted.csv
- 21.crop_predictions.csv
- 22.energy_predictions.csv
- 23.total_exposure_country.csv
- 24.historical_support_flags.csv
- crop_weight_quality_report.csv
- energy_weather_validation.csv

Naming logic

Outputs are numbered in the order they are produced.

Main limitations

- NAHosMIP hosing experiments are idealized stress tests, not probabilistic forecasts.
- The statistical models estimate short-run historical responses, not full adaptation, technological change, trade responses, or market equilibrium.
- Annual energy data measure total household use, not monthly peaks, grid adequacy, or intra-year supply stress.
- Crop-season aggregation cannot distinguish biological growth stages; a frost in April and a frost in June enter the seasonal indicator similarly.
- Fixed baseline crop-area and population weights improve comparability but do not model future relocation of production or households.
- Valuation uses fixed baseline prices and does not estimate welfare losses, substitution, or general-equilibrium feedbacks.
- Scenario indicators outside historical support require cautious interpretation and explicit flagging.

Final contribution

What the thesis delivers

A coherent country-level exposure comparison between standard future warming and severe AMOC weakening for European wheat, barley, household gas, and household electricity.

Why the design is credible

- Country-level outcomes match the available economic data: FAOSTAT and Eurostat.
- Subnational and grid data improve exposure measurement without changing the econometric unit.
- Crop weather is weighted by crop geography; energy weather is weighted by population geography.
- Historical estimation and future scenario construction use the same aggregation rules.
- Validation and historical-support flags make the main methodological risks explicit.

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E-OBS. Daily gridded observations for Europe. Copernicus Climate Data Store / ECA&D.



Eurostat. Complete energy balances, `nrg_bal_c`; electricity and gas prices; annual population.



FAOSTAT. Crops and livestock products; producer prices.

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